

236502 – Artificial Intelligence Project

Mid-Project Update

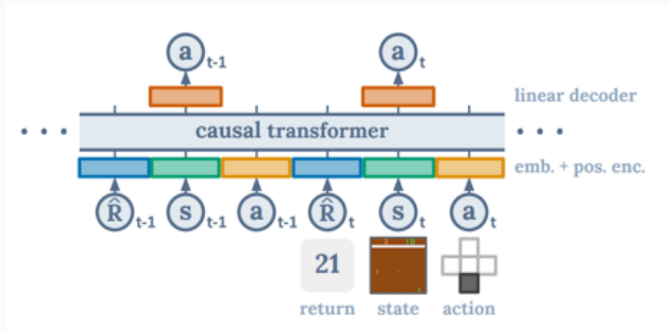
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Project Background

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- **Project Domain** : *Comparison between DT and RL in energy environments.*
- **Key Motivation** : *DT's advantages may improve smart management in multiple-users energy environments.*



Original Project Goals

- **Goal 1 :** *Build and train a DT model in ANM6Easy*
- **Goal 2 :** *Test the model and evaluate performance*
- **Hypothesis :** *The model would satisfying results in the evaluation*

Mid-Project Update

- **Implemented / Experimented :**
 - *Experiment DT building and training on CartPole and HalfCheetah*
 - *Creating robust DT class*
 - *Writing general Trainer class*
 - *Implementing simple evaluation logic*
- **Data Collected :** *Trajectories from random agent in ANM6Easy*
- **Pipelines Built :** *Trajectories collection, DT training, DT evaluation*

Challenges & Adjustments

- **What Didn't Go as Planned :**
 - *Got High loss in the DT training*
 - *Random agents trajectories aren't informative enough*
 - *Searching for a pre-trained smart ANM6Easy RL-agent failed*
- **Adjustments Made :**
 - *Normalized DT input*
 - *Research for PPO trainer in ANM6Easy*

Random DT training

```
Environment created successfully.  
Collecting episode 1/20  
Collecting episode 2/20  
Collecting episode 3/20  
Collecting episode 4/20  
Collecting episode 5/20  
Collecting episode 6/20  
Collecting episode 7/20  
Collecting episode 8/20  
Collecting episode 9/20  
Collecting episode 10/20  
Collecting episode 11/20  
Collecting episode 12/20  
Collecting episode 13/20  
Collecting episode 14/20  
Collecting episode 15/20  
Collecting episode 16/20  
Collecting episode 17/20  
Collecting episode 18/20  
Collecting episode 19/20  
Collecting episode 20/20  
Collected 17 trajectories.  
Model created successfully.  
Epoch: 0 - Loss: 0.7752  
Epoch: 100 - Loss: 0.0836  
Epoch: 200 - Loss: 0.0793  
Epoch: 300 - Loss: 0.0673  
Epoch: 400 - Loss: 0.0479  
Epoch: 500 - Loss: 0.0328  
Epoch: 600 - Loss: 0.0261  
Epoch: 700 - Loss: 0.0210  
Epoch: 800 - Loss: 0.0186  
Epoch: 900 - Loss: 0.0162  
works:)
```


Next Steps

- **Remaining Work :**

- *Build PPO smart agent*
- *Train DT from the smart agent's trajectories*
- *Advance evaluation to multi-objective*
- *Train DT in energynet environment*

- **Timeline to Final Presentation :**

